

LOGICAL POSITIVISM

SYLLABUS ITEM 7, PAPER I SECTION A — Verification Theory of Meaning · Rejection of Metaphysics · Linguistic Theory of Necessary Propositions

ONE MOVE → OWNER FENCE → THREE INHERITANCES → CHRONOLOGY → THE CIRCLE → PHILOSOPHY REASSIGNED → THE TWO-KEY CRITERION → MEANING vs TRUTH → IN PRINCIPLE → STRONG → 1936/1946/1949 → FIVE MOTIVATIONS → SIX STEPS → FOUR PRESUPPOSITIONS → TRACTATUS LEDGER → PSEUDO-STATEMENT PROCEDURE → TWO ROUTES → DAS NICHTS NICHTET → THE CASCADE → AYER vs STEVENSON → EMBEDDING & THE 2020 SENTENCE → NECESSITY TRILEMMA → THE 2024 PAIR → LAWS & HARD CASES → PROTOCOL DEBATE → TOLERANCE → LEDGER, PYQ ROUTING AND CONCLUSION

Read the cyan rail from stage 01 to stage 28. Stages 01–28 are the complete core spine; the single grey stage E is optional enrichment and is never needed for a core answer. Every stage uses a layout chosen for its own content — trees, ladders, chains, matrices, timelines, comparisons, stacks, formulas, numbered rankings and bands.

01

THE ONE MOVE — THREE PRINTED LIMBS, ONE INSTRUMENT

BEFORE 'IS IT TRUE?' ASK 'DOES IT SAY ANYTHING?'

COGNITIVE SIGNIFICANCE

EMPIRICISM ARMED WITH FORMAL LOGIC

PHILOSOPHY = CLARIFICATION

VERDICT FALLS ON THE SENTENCE

THE SINGLE TEST

A sentence makes a **factual** claim only if some **possible experience** would count for or against it. That one test is then turned on metaphysics and on logic, and both results are printed in the syllabus clause.

LIMB 1 — MEANING

Verification theory of meaning

Cognitively meaningful **iff** analytic **or** empirically verifiable

A criterion of **meaning**: never of truth

LIMB 2 — METAPHYSICS

Rejection of metaphysics

Pseudo-statement (Scheinsatz): grammar of an assertion, no logical function

Not false — no truth-conditions, hence no truth-value

LIMB 3 — NECESSITY

Linguistic theory of necessary propositions

Necessity is **verbal and conventional**: tautologies are factually **empty**

No **synthetic a priori** — Kant's category dissolved

ANSWER LINE

Logical positivism is not a set of conclusions about reality but a proposal about **what a philosophical question may be**. Write the limbs in the order **meaning** → **metaphysics** → **necessity** and the answer has a spine before it has content.

02

OWNER SCOPE AND THE FENCE

OWNED HERE

Verification principle: **meaning against truth**: cognitive significance

In-principle verifiability: Ayer's **strong** / **weak** senses

1946 revision: **Church 1949**

Pseudo-statement and the two routes; das Nichts nichtet

The **meaning-knowledge-ontology** cascade

Analytic/synthetic as **exhaustive**: tautology; denial of the synthetic a priori

General statements, laws, theoretical terms

Protocol sentences: tolerance; internal/external

Emotivism as a **corollary**: self-refutation

NOT OWNED — ONE CLAUSE EACH

Falsifiability → POPPER: demarcation, **not meaning**: metaphysics **meaningful**

Two Dogmas, holism, ontological relativity → **TOPIC 11**

Meaning as use, language-games, private language → **TOPIC 08**

Picture theory, saying/showing → **TOPIC 06** — a source, not a version

Hume's epistemology of impressions → **TOPIC 03**

Kant's transcendental machinery → **TOPIC 04**, here only as the target

Eschatological verification (Hick) → **PAPER II**

CORPUS SIGNAL

7 owned parts of 112 in 2018–2025. **2022 is the only year without one**. A low share with a high hit-rate: the printed limbs are compact enough to be tested exactly, so the marks lie in **precision**, not breadth.

03

THE THREE INHERITANCES — AND WHAT WAS NEW IN 1929

CONTENT TEST — HUME

Enquiry **XII**: **relations of ideas** or **matters of fact**

'Commit it then to the flames, for it can contain nothing but sophistry and illusion'

Hume's own target: 'divinity or **school metaphysics**'

→ **TOPIC 03**

FORM TEST — FREGE, RUSSELL

Quantification; the theory of descriptions; Principia Mathematica

Grammatical form ≠ logical form

Lets Carnap show a sentence is **syntactically malformed**

→ **TOPIC 06**

MODEL — EINSTEIN

Relativity forces operational attention to how **distant simultaneity** is fixed and tested

Fix content by fixing the **method of settling**

An **influence**, not a proof that meaning is one measurement

WHAT IS NEW

Hume can say a term lacks an impression. **Carnap can show a sentence is malformed** — a quantifier used as a name. That is a **demonstrable** defect, and the combination of the three inheritances is what distinguishes the Circle from Hume. **Trap**: 'logical positivism is just Hume restated'.

04

CHRONOLOGY RAIL — THE MINIMUM THAT PREVENTS ANACHRONISM

1748

Hume, Enquiry XII — **the fork**

1921

German Logisch-Philosophische Abhandlung

1922

Bilingual Tractatus — never date the English text 1921

1924

The Schlick Circle begins regular meetings in Vienna

1926–27

The Circle reads the Tractatus **aloud**, over a year

1929

MANIFESTO — authors **Hahn, Neurath, Carnap**: dedicated to **Schlick**

1929

Heidegger, Was ist Metaphysik? — das Nichts nichtet

1929–32

Wittgenstein converses with **Schlick and Waismann**; never a member

1930

Schlick, 'The Turning Point in Philosophy'

1932

Carnap, elimination of metaphysics (Erkenntnis)

1932

Neurath, 'Protokollsätze' — **the boat**

1932

Schlick, 'Positivism and Realism' — the **difference-argument**

1932

Carnap, 'Über Protokollsätze' — the form of a protocol is a **choice**

1932–33

Ayer attends Circle meetings as a **visitor**

1933

Schlick, 'Über das Fundament der Erkenntnis' — **affirmations**

1934

Carnap, Logische Syntax (English 1937, \$17) — **tolerance**

1934

Popper, Logik der Forschung — falsifiability; 'official opposition'

1936

Ayer, Language, Truth and Logic — the English transmission

1936

Schlick **murdered**: the Circle disperses; members emigrate

1937 / 1944

Stevenson — Mind paper, then Ethics and Language

1946

LTL 2nd ed. — the revision is in the **Introduction**

1949

Church — the revised criterion admits what it excluded

1950

Carnap, 'Empiricism, Semantics, and Ontology' — internal/external

1951

Quine, 'Two Dogmas of Empiricism' → **TOPIC 11**

1960

Quine, Word and Object — Neurath's boat as **epigraph**

1960–65

Geach sharpens the **embedding** problem against emotivism

FOUR DATE-MARKS

The manifesto's **three authors and dedicatee** · the **1936/1946** split in Ayer · the **1946/1949** gap between revision and refutation · the **non-membership** of Wittgenstein, Ayer and Popper, each for a different reason. None costs more than a clause.

05

THE VIENNA CIRCLE — INSIDE, OUTSIDE, AND THE MANIFESTO

INSIDE THE CIRCLE

MORITZ SCHLICK 1882–1936

RUDOLF CARNAP 1891–1970

OTTO NEURATH 1882–1945

HAHN · WAISMANN

OUTSIDE — AND WHY ASKED

A. J. AYER 1910–89

L. WITTGENSTEIN

KARL POPPER

W. V. O. QUINE

WHAT EACH ONE IS TESTED ON

Leader. The **difference-argument** (1932); 'The Turning Point in Philosophy' (1930); **affirmations** (Konstatierungen) as an incorrigible base (1934)

The logical **elimination of metaphysics** (1932); the **principle of tolerance** (1934/1937 \$17); **internal and external** questions (1950); the 1932 mediation that ended foundationalism

Physicalism: revisable protocol sentences (1932); **the boat**; the encyclopaedic: **unity of science**; anti-foundationalism

Hahn co-authored the manifesto; Waismann recorded the Wittgenstein conversations later published by McGuinness (1979)

THE EXACT REASON

Attended meetings in **1932–33 as a visitor**; wrote the English statement in 1936. **Not a member**. The **strong/weak** terminology is his

Read aloud 1926–27; conversations 1929–32. **Never joined**, and the Tractatus is **not** a verificationist book

Self-described '**official opposition**'. Falsifiability is a criterion of **scientific status**, and he held metaphysics to be **meaningful**

Admired Carnap, then attacked the two pillars: **from inside empiricism** → **TOPIC 11**

THE MANIFESTO, EXACTLY

Wissenschaftliche Weltauffassung; Der Wiener Kreis — 'The Scientific Conception of the World', **1929**. Authors **Hahn, Neurath, Carnap**: **dedicated to Schlick**. A **collective** document announcing a shared attitude and a method, not a list of doctrines every member signed.

06

PHILOSOPHY REASSIGNED, AND THE TWO BASKETS

SCHLICK 1930

'The Turning Point in Philosophy'

Philosophy is **not a system of propositions** beside the sciences

It is an **ACTIVITY** by which the meaning of propositions is made clear

CONSEQUENCE 1

There are **no philosophical propositions** — only clarifications

→ which is why the school's own central claim is later so awkward to classify

CONSEQUENCE 2

Unity of science becomes a project, not a discovery

What unifies knowledge must be a common **language** or **method**

ANALYTIC

True or false **solely** in virtue of the meanings and rules governing its symbols

Necessary · a priori · factually empty

Logic · mathematics · definitions

SYNTHETIC

Contingent · **a posteriori** · **answerable to experience**

Empirical science · history · common sense

May be **false** and still fully meaningful

'COGNITIVE' — THE QUALIFIER

Restricts the claim to the capacity to **state a fact**

A cheer, command, promise, poem, ritual formula or moral exclamation may be intelligible and indispensable **while stating no fact**

INTERNAL DIVERSITY

Three fault lines the examiner can test: the **evidential base** (Schlick against Neurath) · the **language of science** (phenomenalist against physicalist) · the **status of the criterion** (discovery, proposal or rule). '**All positivists held that...**' is almost always false.

07

THE CRITERION — A TWO-KEY LOCK

COGNITIVELY MEANINGFUL <=> ANALYTIC OR EMPIRICALLY VERIFIABLE

failing both -> PSEUDO-STATEMENT (Scheinsatz): no truth-conditions

KEY 1 — ANALYTIC

Wordings: true or false solely in virtue of the meanings of its terms

Yields: necessity, apriority, certainty — and **zero factual content**

'All bachelors are unmarried' · '2 + 2 = 4'

'All objects are either red or not red' → 2024 Q3(b)

KEY 2 — EMPIRICALLY VERIFIABLE

STRONG: conclusively verifiable by a **finite** set of observations

WEAK: some possible observations are **relevant** — would render it probable or improbable

IN PRINCIPLE: the observations can be specified even if no one can make them

'This page is white' → 2024 Q3(b) · 'The earth is flat' — **false, therefore meaningful**

VS

THE BICONDITIONAL

Read **left to right** it **licenses** logic, mathematics and empirical science; read **right to left** it **excludes** everything else. A question that says 'present an exposition' is asking for **both halves**.

08

MEANING AGAINST TRUTH — THE HIGHEST-YIELD DISCRIMINATION

SENTENCE

'The earth is flat'

VERDICT ON MEANING

Meaningful — we know what settles it

VERDICT ON TRUTH

False

THE BICONDITIONAL Read **left to right** it **licenses** logic, mathematics and empirical science, read **right to left** it **excludes** everything else. A question that says 'present an exposition' is asking for **both halves**.

08 → MEANING AGAINST TRUTH — THE HIGHEST-YIELD DISCRIMINATION

SENTENCE	VERDICT ON MEANING	VERDICT ON TRUTH
'The earth is flat'	Meaningful — we know what settles it	False
'There are mountains on the far side of the moon' (1935)	Meaningful — verifiable in principle	Later established true
'This page is white'	Meaningful — synthetic, checkable	True or false by inspection
'All objects are either red or not red'	Meaningful — analytic, $P \sqcup \neg P$	True, and factually empty
'The Absolute is beyond time'	Not cognitively meaningful	No truth-value assigned
THE CONSEQUENCE Because the metaphysical sentence receives no truth-value , the positivist may not add that the Absolute does not exist — a denial would require the very truth-conditions he says are missing. The rejection is at the level of the sentence, not of the object.		
MEMORY False means the world said no. Meaningless means the world was never asked.		

09 → IN PRINCIPLE AGAINST IN PRACTICE — AND WHAT THE VERDICT EXCLUDES

THE QUALIFIER THAT SAVES ASTRONOMY
In principle: the settling observations can be **specified** → bears on **meaning**
In practice: present technology, funding, location → **irrelevant**
'Mountains on the far side of the moon' was meaningful long before spaceflight
'The Absolute is beyond **all** experience' fails **even in principle**, because the sentence itself forbids the specification

VS

WHAT 'COGNITIVELY MEANINGLESS' DOES NOT SAY
Not that the words are unintelligible
Not that the utterance is worthless
Not that expressive, ritual, practical or poetic uses are abolished
Not that the speaker is insincere or confused
It asserts **only** that the sentence does not **state a fact**

pseudo-statement (' <i>Scheinsatz</i> ') category: grammatical form of a statement without its logical function	
cognitively meaningless	The verdict: no truth-conditions, hence no truth-value
'false'	Wrong — falsity presupposes the truth-conditions the positivist says are absent
'unprovable'	Wrong — this concedes that a proposition has been expressed
'nonsense', unqualified	Risky — it invites the ordinary-language reading; use only with the technical gloss

10 → THE STRONG SENSE AND ITS CASUALTIES

CONCLUSIVELY ESTABLISH **FINITE SET OF OBSERVATIONS** **RIGOROUS — EXCLUDES METAPHYSICS CLEANLY** **AND EXCLUDES SCIENCE**

- 1** **Universal laws.** 'All metals expand when heated' quantifies over **indefinitely many** instances; no finite observation set exhausts them, so the sentence is not verifiable and hence not meaningful
- 2** **Statements about the past.** 'Caesar crossed the Rubicon' cannot be observed now; present records supply only the **indirect** relation the strong sense forbids
- 3** **Statements about other minds.** Another's pain is not an object of my observation, so either the sentence is meaningless or it must be reduced to observed behaviour — a **specific behaviourist commitment**, not a neutral consequence of empiricism
- 4** **What survives:** singular present-tense observation reports — 'this page is white'

VERDICT **Strong verification kills science.** A criterion of meaning that convicts the sentences of physics is self-evidently the wrong criterion, and **Ayer says so**. Naming only universal laws gives one third of the casualty list and weakens the dilemma.

11 → THE WEAK SENSE AND ITS COST

SOME OBSERVATION RELEVANT **PROBABLE OR IMPROBABLE** **SAVES LAWS, THE PAST, OTHER MINDS** **RELEVANCE IS CHEAP**

- 1** Weak verifiability asks only that some possible observation be **relevant to** the statement's truth or falsehood
- 2** **Conjoin** the metaphysical sentence with an **auxiliary hypothesis**, and an observation-statement follows from the **pair** that does not follow from the auxiliary **alone**
- 3** If that suffices for significance, 'the Absolute is perfect' can be given a passport by any sufficiently ingenious conjunction — **the criterion stops excluding**
- 4** **This is the exact structure of Church's 1949 objection** against the 1946 repair

THE FORK Tight enough to exclude metaphysics is tight enough to exclude physics; loose enough to admit physics is loose enough to admit metaphysics. **No setting of the dial does both jobs**. This is the principle's **internal** problem, and it is **distinct** from self-refutation at stage 28.

ATTRIBUTION FENCE **Strong/weak is Ayer's vocabulary.** Do not write 'the Vienna Circle distinguished strong and weak verification' as though it were one agreed Circle doctrine.

12 → THE REPAIR AND ITS FAILURE — 1936 · 1946 · 1949

1936 — LTL, 1st ed.
The **weak** test adopted
Observations that make the proposition **probable**
Vague, and Ayer knew it

→

1946 — LTL, 2nd ed., INTRODUCTION
Directly verifiable: it is an observation-statement, or with observation-statements it entails one **not deducible from those premisses alone**
Indirectly verifiable: with further premisses — analytic, directly verifiable, or independently indirectly verifiable — it entails a directly verifiable statement not deducible from those premisses alone

→

1949 — CHURCH
The revised criterion **admits statements it was designed to exclude**
Given three mutually independent observation-statements, an **arbitrary** sentence satisfies the conditions

MIGRATION **Verification — confirmation and testability of systems. Gain:** a far better account of how evidence supports scientific theory. **Loss:** the sharp, sentence-by-sentence boundary between the meaningful and the meaningless. **Never merge 1946 and 1949.**

13 → THE POSITIVE CASE — FIVE MOTIVATIONS, GRADED

- 1** **DIFFERENCE-ARGUMENT****Schlick, 'Positivism and Realism', 1932.** If two hypotheses agree on **every** possible observation they do not differ in **content** — they are one hypothesis in two vocabularies. A principle of **significance**, not of epistemology, which is why it outlives the criterion built on it as a weaker heuristic.
STRONGEST — AND IT SURVIVES
- 2** **THE EINSTEIN MODEL**relativity forces operational attention to how claims such as **distant simultaneity** are fixed and tested. **Limit:** not a proof that every concept reduces to one measurement procedure; strict operationalism makes the same concept measured two ways into two concepts.
STRONG INFLUENCE
- 3** **MEANING AND USE-CONDITIONS**; a factual sentence includes knowing circumstances relevant to asserting and testing it. **Caution:** do **not** import later meaning-as-use. The Circle builds **one uniform empiricist test**; the later Wittgenstein refuses uniformity itself → **TOPIC 08**.
QUALIFIED
- 4** **THE DEMARCATION PROBLEM** criterion separating empirical science from metaphysics. **Popper's rival:** falsifiability is a criterion of **scientific status**, he held metaphysics to be **meaningful**, and he was **not a member**. Different question, not a repair.
CONTESTED
- 5** **UNITY OF SCIENCE** Phenomenalist reduction, and later physicalist public-language projects, sought intertranslation or coordination among the sciences. **Carnap and Neurath differed** over reduction, physicalism and encyclopaedic unity → stage 27.
INTERNALLY DISPUTED

ANSWER-CRAFT Open a 'discuss the verification theory' answer **here**, not with self-refutation. A script that first makes the doctrine attractive and then dismantles it demonstrates the control that fifteen and twenty marks are for.

14 → THE ARGUMENT IN SIX STEPS

- 1** To **understand** a sentence is to know what it would be for it to be **true**
- 2** To know that is to **distinguish** the circumstances in which it holds from those in which it does not
- 3** To distinguish those circumstances is to know what would count as: **evidence either way**
- 4** ∴ understanding a sentence **is** grasping its **verification-conditions**
- 5** ∴ a sentence for which **no evidence could count either way** has **nothing for understanding to consist in**
- 6** ∴ such a sentence is **not false**; it is **without cognitive content**

15 → THE FOUR PRESUPPOSITIONS — AND THE OBJECTION MAP THEY GENERATE

#	THE UNDEFENDED ASSUMPTION	THE OBJECTION IT GENERATES
P1	Understanding is exhausted by truth-conditions and truth-conditions are exhausted by observation-conditions	The identification is made silently at step 2–3 and is never argued for
P2	Observation-statements are epistemically privileged — there is a level of report not itself theory-laden	Attacked from within by the protocol debate of 1932–35 → stage 26
P3	Sentences face experience individually	Duhem-Quine holism — TOPIC 11 — and already strained by the positivists' own theory-network account of theoretical terms → stage 25
P4	The analytic/synthetic distinction is sharp — the criterion's first disjunct depends on it	'Two Dogmas' (1951) attacks the boundary rather than settling it → TOPIC 11
WHY THIS EARNS 20-MARK SPACE It converts an unordered list of objections into a structure : every major criticism is the denial of one hinge , and the script can say which hinge and why		

16 → THE TRACTATUS LEDGER — WHAT WAS TAKEN, WHAT WAS ADDED

TAKEN FROM THE TRACTATUS
Sense = **truth-conditions** (4.022, 4.024)

ADDED BY THE CIRCLE
Meaning = **verification-conditions**

P4

The analytic/synthetic distinction is sharp — the criterion's **first disjunct** depends on it

Two Dogmas' (1951) attacks the boundary rather than settling it → TOPIC 11

WHY THIS EARNS 20-MARK SPACE

It converts an unordered list of objections into a **structure**: every major criticism is the **denial of one hinge**, and the script can say **which hinge and why**

16

THE TRACTATUS LEDGER — WHAT WAS TAKEN, WHAT WAS ADDED

TAKEN FROM THE TRACTATUS

Sense = **truth-conditions** (4.022, 4.024)

Bipolarity is a **logical** requirement: a determinate place in logical space

Metaphysics is nonsensical (unsinnig) because it fails to **depict**

The base is **objects never exemplified** — a **logical postulate**

Value and the mystical **manifest** but are not factual propositions

Also: the boundary project · logic as **tautology** · philosophy as an **activity** · the exposure of surface grammar

ADDED BY THE CIRCLE

Meaning = **verification-conditions**

Verifiability is an **epistemic** requirement: **someone must be able to check**

Metaphysics is meaningless because **no observation is relevant**

The base is **observations / protocols** — an **empirical** claim, and therefore **disputable**

Ayer gives value utterances an **emotive** role — an added metaethical theory

VS

THE CONSEQUENCE

Universal laws have **specifiable truth-conditions** but **no finite conclusive verification-conditions** — so the substitution **itself** generates the strong/weak dilemma

FENCE (i)

Wittgenstein was **never a member** of the Vienna Circle

FENCE (ii)

The Tractatus is **not a verificationist text**: the substitution is the **Circle's own** empiricist addition

FENCE (iii)

Later Wittgenstein endorses **no** single criterion of meaning at all — **TOPIC 08**

17

THE PSEUDO-STATEMENT IDENTIFICATION PROCEDURE

1

Is the sentence **ANALYTIC** — true or false by the meanings and rules of its terms alone? **Yes** → **meaningful and factually empty. Stop.**

2

Can any possible observation count for or against it, in principle? **Yes** → **meaningful and synthetic. Stop.**

3

Diagnose the route. ROUTE 1: a constituent **term** has no empirical criterion of application. ROUTE 2: individually significant terms are **combined** so that no truth-conditions result

4

Verdict: **pseudo-statement (Scheinsatz)**. Record what is **not** claimed — not false, not unproven, not unintelligible; expressive and practical force untouched

SOURCE

Carnap: 'Überwindung der Metaphysik durch logische Analyse der Sprache', Erkenntnis, 1932. Ayer: a metaphysical sentence 'purports to express a genuine proposition, but does not'. **2021 Q3(b) asks for this procedure**, not for a list of condemned sentences.

18

THE TWO ROUTES, WORKED — WITH CONTROLS

SENTENCE

'The soul is immortal'

'The Good transcends Being'

'Caesar is a prime number'

'The nothing nothings' (das Nichts nichtet)

'Electrons have negative charge'

'2 + 2 = 4'

ROUTE

ROUTE 1

ROUTE 1 + 2

ROUTE 2

ROUTE 2 — the paradigm

CONTROL — MEANINGFUL

CONTROL — MEANINGFUL

DIAGNOSIS

No empirical criterion of application for 'soul'; no observation bears on the predicate

Neither term has a criterion, **and** the relation yields no truth-conditions

Grammatical, every word significant — and **no verification procedure applies** to the combination

A **double violation**: a logical quantifier used as a **name**, and a **pseudo-verb** predicated of it → stage 19

Empirically testable, though **indirectly**, through a predictive theory-network → stage 25

Analytic; true by the rules of arithmetic: **factually empty** → stage 23

WHY BOTH ROUTES

Route 1 alone cannot handle 'Caesar is a prime number'; route 2 alone cannot handle 'the soul is immortal'. **A single route is half a procedure**, and without a **control case** the test reads as a blacklist rather than a diagnostic.

19

'THE NOTHING NOTHINGS' — DIAGNOSIS, REPLY AND RENDERING RULE

SOURCE — HEIDEGGER, 1929

Was ist Metaphysik?: 'Das Nichts nichtet'

ANALYSIS — CARNAP, 1932

A **double violation**: (a) 'nothing' is a **logical quantifier** used as if it **named** something; (b) 'nothings' is a **pseudo-verb** predicated of that pseudo-name. The sentence is parasitic on the form 'Something does X'. The defect is in **logical syntax**, not in ordinary grammar, and is therefore **demonstrable**

RENDERING RULE

Correct: 'Nothing noths' or 'the nothing nothings'. **Forbidden**: 'Nothing noughts' is **not** a rendering of this phrase and must never be written

HEIDEGGER'S AVAILABLE REPLY

Nichts is disclosed in the mood of **anxiety** (Angst), and the breach of logical grammar is **deliberate**, since **logic itself presupposes** the prior disclosure of beings as a whole. **Report it, then rule** — otherwise the answer is ridicule, not assessment

ATTRIBUTION

The **sentence** is Heidegger's; the **analysis** is Carnap's. Never merge them

20

THE CASCADE — MEANING, KNOWLEDGE, NATURE OF THINGS

LEVEL

MEANING — the root

KNOWLEDGE

NATURE OF THINGS

WHAT THE POSITIVIST SAYS

The sentence has **no truth-conditions**: it is a **pseudo-statement**

The question is **dissolved**, not answered: one can neither know nor fail to know what is not even a proposition

The **question** of such entities' existence is **ill-formed**

WHAT HE MAY NOT SAY

—

That we are **ignorant** of metaphysical truths — agnosticism presupposes a proposition whose truth-value is undetermined

That the Absolute, the soul or God **does not exist** — a denial requires the very truth-conditions declared absent, which is also why Ayer's position is **not atheism**

THE CONNECTION

Meaning grounds knowledge, and knowledge grounds claims about the world. Cutting at the level of **meaning** simultaneously eliminates **pseudo-knowledge** and **pseudo-ontology**

2023 Q3(a) VERDICT

All three, through meaning as the root. Saying 'all of them together' **without the ordering** describes the question instead of answering it.

21

ETHICS AND AESTHETICS — AYER AGAINST STEVENSON

AYER — LTL 1936, ch. 6

Pure expression of feeling: adding '...is wrong' to 'you stole that money' adds **no factual content**; it **evinces** disapproval and may **evoke** it

No propositional paraphrase at all

Ethical disputes are **always really about facts**: once all facts are agreed, 'we should not attempt to argue'

Moral argument: **essentially impossible** except about the surrounding facts

Emotivism is a **corollary of the verification principle** — derived **top-down** from a theory of meaning

STEVENSON — Mind 1937; Ethics and Language 1944

Two kinds of meaning: a **descriptive** meaning varying with context and an **emotive** meaning — a disposition built by usage to express and evoke attitude

Model: 'I approve of X; do so as well' — expression **plus** an imperative element

Disagreement in belief against disagreement in attitude: the latter **can survive complete factual agreement**

Moral argument is possible: reasons work **causally**, by altering the beliefs on which attitudes rest — **rational** against **persuasive** methods

Persuasive definition: 'real democracy is...'. Built **bottom-up** and **independent** of the criterion

VS

SHARED — AND WORTH A MARK BY ITSELF

Both are **non-cognitivists, not subjectivists**. Neither treats 'X is wrong' as a **report** of the speaker's feeling, because that would turn moral disagreement into disagreement about **autobiography**. **Expression is not report.**

EXECUTABLE VERDICT

Ayer's emotivism is a **deduction** from the criterion and shares its fate; Stevenson's is an **independent theory of linguistic influence** which explains what Ayer could not — how moral argument can be genuine though moral judgements state no fact.

22

THE EMBEDDING PROBLEM, THEOLOGY AND THE 2020 SENTENCE

FREGE-GEACH EMBEDDING

Geach 1960, 1965; anticipated by Frege. 'If lying is wrong, then getting your brother to lie is wrong.' The **antecedent is unasserted** and expresses **no attitude**, yet it must mean the **same** as the free-standing 'lying is wrong' for the modus ponens to be valid. **Decisive** against simple emotivism; Blackburn and Gibbard later construct logics of attitude — **neither Ayer nor Stevenson had an answer**

DISAPPROVAL vs DISTASTE

Broccoli against murder. **Stevenson** can appeal to the **generality** and **social function** of moral attitudes: **Ayer cannot**

MORAL PROGRESS

Stevenson: progress is convergence of attitude under **fuller belief**. **Ayer** concedes the point

THEOLOGY — OWNED DEPTH

Transcendent existence-claims fail by **route 1. This is not atheism** — atheism is a **meaningful denial**; the positivist declines the question. Ritual, expressive and communal uses are a separate question. **Hick's eschatological verification** is one clause → **PAPER II**

2020 Q1(e) — THE SPLIT IS THE SUBJECT

The Soul of Superman' — no empirical criterion of application, fictional bearer (**route 1**). **Predicate**: 'is Good' — adds no factual content; **evinces** approval (**emotive**). **Whole**: neither analytic nor empirically verifiable — **cognitively meaningless, not false**. **Qualify**: within an explicitly stated **fiction-framework**, 'in the story, Superman is portrayed as good' **is** checkable

23

NECESSITY MADE VERBAL — THE TRILEMMA AND THREE RIVALS

THE TRILEMMA

Logic and mathematics are **necessary + a priori + certain**, yet empiricism holds that **all knowledge comes from experience**. State this **before** giving the answer, or the doctrine arrives as vocabulary rather than as a solution.

KANT — SYNTHETIC A PRIORI

Necessary truths are **synthetic** — they say something about the world — yet known **a priori**, because the mind **imposes** spatio-temporal and categorical structure

Problem: empiricists reject a **third box** beyond analytic and synthetic

MILL — EMPIRICAL GENERALISATION

'2 + 2 + 4' is a very well-confirmed **empirical generalisation** about collections of objects

Problem: absurd — we do not **discover** it by counting pebbles, and **no experience could refute it**: an apparent counter-instance is put down to **miscounting**

POSITIVISM — LINGUISTIC

Necessary truths are **analytic**: true solely in virtue of the **meanings and conventions** governing the symbols

They are **tautologies** and say **nothing about the world**

Their necessity is **verbal**

FEATURE

NECESSARY

A PRIORI

CERTAIN

EMPTY

HOW THE POSITIVIST ACCOUNT DELIVERS IT FROM ONE SOURCE

Their truth follows from the rules laid down for the symbols; **no experience could refute them because they say nothing experience could bear on**

Grasping the meanings of the terms **suffices** for knowing their truth; no observation is needed

To deny them is to **violate one's own conventions** — self-contradiction

'All bachelors are unmarried' records our **resolve** to use two expressions interchangeably — **the price, and the point**

MEMORY

A necessary proposition is **certain because it forbids nothing**

24

THE EXHAUSTIVE DICHOTOMY AND THE 2024 PAIR

'ALL OBJECTS ARE EITHER RED OR NOT RED'

Analytic (tautology) form: **P ∨ ¬P**

'THIS PAGE IS WHITE'

Synthetic (empirical)

24

THE EXHAUSTIVE DICHOTOMY AND THE 2024 PAIR

'ALL OBJECTS ARE EITHER RED OR NOT RED'

Analytic / tautology, form $P \sqcup \neg P$

Meaningful as a logical truth: true under all truth-conditions

Says **nothing** about the world; **no observation could refute it because it is empty**

Its truth depends only on the meanings of 'or' and 'not'

VS

'THIS PAGE IS WHITE'

Synthetic / empirical

Meaningful because verifiable: look at the page

Its truth depends on an **observable fact**

It **could be false** — the page might be blue

THE ANSWER AS ASKED NO, Both are meaningful, but by **different branches** of the criterion. Metaphysics — 'the Absolute is beyond time' — fits **neither**. There is **no third box**, so the **synthetic a priori** is **declared incoherent** and Kant appears here **only as the target** — TOPIC 04.

QUINE 1936 — "Truth by Convention"ions must be **applied**, and deriving their infinitely many consequences **already presupposes logic** — circularity — TOPIC 11

QUINE 1951 — "Two Dogmas"Analyticity cannot be **non-circularly defined**, so the linguistic theory rests on a category that is itself unclear

APPLICABILITY If mathematics is empty, why is it **indispensible** in physics? **Reply**: applicability belongs to the **embedding empirical theory**, which supplies the contact with observation

GÖDEL Limitative results constrain **particular formal systems**; they are **not** a one-line refutation of every conventionalist account — do not overreach

25

GENERAL STATEMENTS AND THE FOUR HARD CASES

LAW + CONDITIONS + AUXILIARIES

'All metals expand when heated' cannot be **conclusively** verified

It reaches experience only **in company**

→

PREDICTION

'This copper bar, heated, will lengthen'

A **testable consequence** of the whole system

→

OBSERVATION REPORT

Match → the system is **confirmed** (supported, never proved)

Fail → the **system** is revised; the law is one candidate among several

HARD CASE	THE ROUTE	THE RESIDUAL COST
Universal laws	Law + initial conditions + auxiliaries → prediction → observation; later confirmation theory supplies degrees of support	Support is never proof, and failure never points at one sentence — which is where holism later enters
The past	Present records, traces and testimony , read through causal and historical theory	A strict phenomenalist reduction confuses the evidence with the content — the sentence is about Caesar, not about the parchment
Other minds	Publicly observable behaviour, language and explanatory theory	Reducing pain simply to behaviour is one behaviourist route, not the only possible weak-verification account
Theoretical entities	Significance through the term's role in laws, auxiliaries and the testable consequences of a theory-network	Network content means meaning is not assigned sentence by sentence — and the criterion was a sentence-by-sentence criterion

2019 Q2(a) — THE EXTENSION REFUSED 'All reality is spiritual' generates **no prediction under any initial conditions and any auxiliaries**. Scientific universals have **systematic empirical consequences**; metaphysical universals specify **no comparable test**. The difference is **categorical**, not one of degree.

2018 Q3(b) — THE HONEST RESIDUE Not on the strong sense; on the weak sense **only at the cost of the sharp boundary**. Fence: potential **falsification** is Popper's **demarcation** proposal, not Ayer's definition of meaning.

26

THE PROTOCOL-SENTENCE DEBATE — THE CRISIS INSIDE THE CIRCLE, 1932–35

SCHLICK 1934 — FOUNDATIONALISM

'Über das Fundament der Erkenntnis'

Base: Konstatierungen — **affirmations** of the form 'here now blue'

Status: **absolutely certain**, because understanding the sentence and verifying it are **one and the same act** — no gap, for error to enter

Reality: the '**unshakable point of contact**' between the system of knowledge and reality

The catch: an affirmation **cannot be written down and preserved** — recording it yields a **hypothesis about a past moment**. Its certainty is **momentary and unusable**

VS

NEURATH 1932 — COHERENTISM + PHYSICALISM

'Protokollsätze'

Base: **physicalist**, fully explicit protocols **including the observer and the time**

Status: **fallible and revisable** like any other sentence; **nothing is sacrosanct**, and even a protocol may be **struck out** if it conflicts with the system

Reality: statements are compared with statements, **not with "experiences"** — no exit from language to an unmediated given

The image — the boat: 'We are like sailors who have to rebuild their ship on the open sea, without ever being able to dismantle it in dry-dock and reconstruct it from the best components'

CARNAP'S MEDIATION, 1932'Über Protokollsätze': the **form** of protocol sentences is **not a fact to be discovered** but a **matter of choice of language** — tolerance applied to the Circle's **own** foundations. **This effectively ends foundationalism inside the Circle**

SCHLICK'S OBJECTION If truth is **coherence** with the accepted system, then a **consistent fairy-tale** is as 'true' as science; coherence **alone** cannot select between rival internally consistent systems

NEURATH'S REPLY The selection is made not by **logic** but by the **actual encyclopaedic practice** of scientists, constrained by the protocols they in fact accept

RESIDUAL — WHERE THE MARKS ARE: the choice **sociological rather than epistemic**. Schlick's charge is **not fully met**, and Neurath accepts a naturalistic consequence Schlick found unacceptable

CASUALTY AND AFTERMATHPresupposition **P2** — theory-neutral observation — is broken **from within**. Quine uses the boat as the **epigraph** of Word and Object (1960): an acknowledged **affinity**, not a derivation — TOPIC 11

27

TOLERANCE, FRAMEWORKS, REDUCTION AND THE UNITY OF SCIENCE

IN LOGIC, THERE ARE NO MORALS

give SYNTACTICAL RULES instead of PHILOSOPHICAL ARGUMENTS

Logische Syntax der Sprache 1934 | The Logical Syntax of Language 1937, s.17

THE FOUR STEPS

1 There is **no single 'correct' logic or language given by reality** — only **linguistic frameworks**, each constituted by its own rules. 2 A framework is chosen on **pragmatic** grounds — fruitfulness, simplicity, convenience — **not** on grounds of truth. 3 ∴ traditional metaphysical disputes are **disguised proposals** about which framework to adopt. 4 ∴ intuitionist against classical logic, or phenomenalist against physicalist language, is a **choice of instrument**.

INTERNAL QUESTION

'Is there a prime number greater than 100?' asked **within** the arithmetical framework.

Settled by **mathematical rules and proof**. Yes

Ontology survives as internal employment

EXTERNAL QUESTION

'Are there **really** numbers?' asked **about** the framework as a whole

Not a theoretical question at all — the **practical** question whether to adopt the number-language, answered by **usefulness**

Metaphysics is **reclassified**, not merely condemned

WHY IT IS STRONGER THAN AYER'S

It explains why metaphysical disputes are **interminable** — proposals have no truth-values

It leaves **internal ontology legitimate**

It replaces a controversial criterion with a **checkable demand** for explicit rules

PROGRAMME	CORE CLAIM	CAUTION
Phenomenalist / reconstructionist	Scientific and object talk is reconstructed from elementary experiential statements	Risks private-language and other-minds problems
Physicalist / public-language	Protocols and scientific claims use an intersubjectively testable physical language	Neurath's and Carnap's versions differ; it is not simple reduction to private sensation
Unity of science	The sciences coordinated in a common language or encyclopedia	'Unity' may mean translation, coordination or methodological integration — not one settled reduction theorem

QUINE'S BOUNDED ATTACK — AND THE REPLY Quine holds that the internal/external distinction depends on an **analytic/synthetic boundary he rejects**, favouring ontology continuous with total theory. **Carnapian reply**: pragmatic framework choice need **not** be a disputed analytic truth. **Verdict**: tolerance **refines** anti-metaphysics; Quine **pressures**, but does not by assertion **settle** the boundary — TOPIC 11

28

THE CRITICISM LEDGER, PYQ ROUTING, ANSWER SPINE AND CONCLUSION

PRESSURE — RANKED	THE OBJECTION	REPLY AND RESIDUE
1 STRONG/WEAK FORK INTERNAL, decisive	Strong verification excludes universal laws, the past and other minds — kills science . Weak verification is so permissive that an auxiliary hypothesis readmits metaphysics	Ayer's 1946 direct/indirect machinery. Residue : Church 1949 — the repair admits what it excluded. No formulation ever worked
2 PROTOCOL DEBATE INTERNAL, structural	The school could not agree what verification terminates in	Carnap 1932: the form of a protocol is a matter of choice . Residue : a chosen base cannot ground a criterion advertised as a discovery about meaning
3 ANALYTICITY + HOLISM EXTERNAL, bounded	Quine 1951: (i) the analytic/synthetic distinction cannot be non-circularly defined, so the linguistic theory rests on an unclear category; (ii) reductionism is false — statements face experience as a corporate body	Carnapian pragmatic replies remain available. Residue : if Quine succeeds, the two-basket programme loses central supports — TOPIC 11
4 SELF-REFUTATION CLASSIC, over-rated	The principle is neither analytic (its denial is coherent) nor empirically verifiable (no observation tests it) — meaningless by its own standard	(i) a recommendation — cost: a policy may be declined . (ii) analytic of the proposed concept — cost: circular . (iii) a meta-linguistic rule — cost: the question recurs

CONTRASTS, NOT REPAIRS

POPPER — **demarcation**, not meaning; metaphysics **meaningful**; 'official opposition'

LATER WITTGENSTEIN — rejects **one essence** of meaning; varied uses in language-games; positive theory — TOPIC 08

ORDINARY LANGUAGE — commands, avowals, performatives, jokes, mathematics and religious practices may not be **falsified empirical reports** at all; this does **not** vindicate metaphysics

PYQ ROUTING — 7 OF 112

2018 Q3(b) 15 — how shown meaningless + **all** scientific sentences? → 17 · 18 · 25

2019 Q2(a) 20 — general statements + extension? → 25, both halves

2020 Q1(e) 10 — 'The Soul of Superman is Good' → 22, the **split**

2021 Q2(b) 15 — pseudostatements: what, how, examples → 17 · 18 · 19

2023 Q3(a) 20 — meaning / knowledge / nature of things? → 20, then 27 · 28

2024 Q3(b) 15 — tautology against observation → 24, **answer no**

2025 Q3(b) 15 — exposition + strong/weak → 7 · 10 · 11 · 12

2022 is the **only year without an owned part**. Supporting, owned elsewhere: 2019 Q1(e) and 2024 Q4(c) → TOPIC 11; 2025 Q3(a) → TOPIC 08

ANSWER SPINE AND DIRECTIVES

10 marks: criterion → one **false-but-meaningful** illustration → the verdict word → one qualification

15 marks: frame (**meaning not truth**) → **both halves** of the biconditional → the doctrine asked for → one named objection with its reply → graded verdict

20 marks: frame → full doctrine → the hard cases → **two** of the four pressures, ranked, at least one **internal** → the doctrine's own **development** (Carnap after Ayer) → qualified conclusion

Directives: exposition demands: **both halves** 'differentiate' demands the **dilemma** 'how does one identify' demands a **procedure** 'critically' demands the opponent's **reply** before the ruling 'all of them together' demands an **ordering**

HISTORICAL END Schlick was **murdered in Vienna in 1936**, and the Circle dispersed under political pressure, its members emigrating. The **institutional** and the **philosophical** ends are distinct, and the dissolution continued in Britain and America afterwards.

QUALIFIED CONCLUSION Logical positivism proposed a **single test** to separate what can be said from what only looks as though it were said, and the test failed in **four different ways** — **two of them discovered by positivists themselves**. What survives is not the doctrine but a **discipline**. Schlick's **difference-argument** as a heuristic of significance, the demand for **explicit rules** rather than obscurity, the permanent separation of criteria of **meaning** of **truth** and of **scientific status**, and an agenda — confirmation, demarcation, holism, analyticity — that is simply the list of problems the Circle could not solve. **State the achievement and its matching cost in the same sentence.**

E

OPTIONAL ENRICHMENT — NOT REQUIRED FOR A CORE ANSWER

Everything above is the complete core spine, stages 01–28. Nothing in this single subordinate stage is needed to answer any of the **seven** routed questions; use it for **one closing clause** at most.

OPTIONAL MODULE	THE SECOND-ORDER ISSUE	SAFE USE / PROMOTION TRIGGER
A1 Repair or abandonment?	Granting the core narrative, was Neurath's coherentism a repair of empiricism or an abandonment of it, and does 'intersubjective testability' still deserve the name verification ?	One closing evaluative clause. The debate itself is already core at stage 26; re-promote only if a question asks the coherence issue in its own right.
A2 Carnap against Neurath on physicalism	Should all meaningful statements be translated into a physical language, or merely coordinated inside a publicly testable encyclopedia?	One extra sentence when assessing whether positivism was more nuanced than the slogan 'metaphysics is nonsense'. Promote if a question asks about physicalism or unity of science.
A3 From verification to confirmation	Did later positivism preserve its meaning-criterion, or retreat into a weaker philosophy of scientific support?	The last paragraph only. Do not replace the verification principle with Hempelian confirmation theory in the main answer.

SOURCE TRAIL Moritz Schlick · Rudolf Carnap · Otto Neurath · A. J. Ayer, Language, Truth and Logic · Carl Hempel · Alonzo Church · Michael Friedman · Thomas Uebel.